Faithful Character-Torus Recovery for a Scaled Graph-Directed Hénon Hardy Owner

Route-A structural certificate C134

Abstract

We replace a single finite character of graph-directed affine Hénon branch translations by their labelled character torus. On a uniformly separated scaled family, the same trace-class Hardy operator has an all-order trace law, Fredholm lattice product, and primitive-cycle expansion. Its first three normalized logarithmic jets recover every branch-labelled integer translation exactly. A Gaussian-rational faithful character distinguishes two models that alias under every fifth-root phase. Recovery is exact only in this frozen labelled family; it is neither a stability theorem nor arbitrary geometric or target recovery.

1 Scaled separated family

Fix

$$A = \begin{pmatrix} 3/16 & -1/32 \\ 1/4 & 0 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}, \quad c = (1/2, 1/3, 1/5).$$

For an integer $k \geq 1$, let $t = (t_0, t_1, t_2)$ be any branch permutation of $(-2k, 0, 2k)$ and set $\phi_j(z) = Az + (t_j, 0)$ on three copies of $H^2(\mathbb{D}^2_{3k})$. The coordinate image radii are $21k/32$ and $3k/4$. The first-coordinate interior margin is $11k/32$, and adjacent images have gap $11k/16$. Thus every branch is compactly contained and the three images are pairwise disjoint, uniformly after scaling.

Every admissible cyclic word has one affine fixed point. Strong separation makes its itinerary unique, so primitive admissible necklaces biject with primitive geometric cycles at all periods. The exact replay prefix is

period	1	2	3	4	5	6	7	8
rooted closed words	1	3	7	11	21	39	71	131
primitive cycles	1	1	2	2	4	5	10	15

2 One global character-family owner

For a labelled $u \in U(1)$ put

$$W_{t,u} = B \operatorname{diag}(c_j u^{t_j}), \qquad (\mathcal{L}_{t,u} f)_i(z) = \sum_j B_{ij} c_j u^{t_j} f_j(\phi_j(z)).$$

Compact interior restriction and the total-degree multiplicity $m+1$ give a summable trace-class majorant, uniformly in u. Since the eigenvalues of A are $1/8, 1/16$, triangularity by polynomial degree yields, for every $n \geq 1$,

$$\operatorname{Tr} \mathcal{L}^n_{t,u} = \frac{\operatorname{Tr} W^n_{t,u}}{(1 - 8^{-n})(1 - 16^{-n})}. \tag{1}$$

Consequently the entire Fredholm determinant has the normally convergent lattice product

$$D_{t,u}(z) = \det(I - z\mathcal{L}_{t,u}) = \prod_{r,s \geq 0} \det(I - z 8^{-r} 16^{-s} W_{t,u}), \tag{2}$$

and grouping rooted words by primitive root gives

$$\log D_{t,u}(z) = -\sum_{[\gamma]} \sum_{m \geq 1} \frac{(c_\gamma u^{M_\gamma} z^{\ell_\gamma})^m}{m \det(I - A^{m\ell_\gamma})}.$$

These are all-period identities; period eight is only finite replay.

3 Three logarithmic jets recover translations

Work first in $\mathbb{Q}[X, X^{-1}]$. Direct expansion gives

$$\det(I - zW_{t,X}) = 1 - \tfrac{1}{2}X^{t_0}z - \tfrac{1}{6}X^{t_0+t_1}z^2 - \tfrac{1}{30}X^{t_0+t_1+t_2}z^3. \tag{3}$$

Define the normalized log jets

$$P_n = -n(1 - 8^{-n})(1 - 16^{-n})[z^n]\log D_{t,X}.$$

Equation (1) gives $P_n = \mathrm{Tr}(W_{t,X}^n)$. Newton identities then recover

$$E_1 = P_1, \quad E_2 = \frac{P_1^2 - P_2}{2}, \quad E_3 = \frac{P_1^3 - 3P_1P_2 + 2P_3}{6},$$

and (3) gives

$$2E_1 = X^{t_0}, \qquad -6E_2 = X^{t_0+t_1}, \qquad 30E_3 = X^{t_0+t_1+t_2}.$$

If the three exponents are S_0, S_{01}, S_{012}, then

$$t_0 = S_0, \qquad t_1 = S_{01} - S_0, \qquad t_2 = S_{012} - S_{01}.$$

Thus three labelled universal log jets determine the complete integer triple.

4 Exact alias and faithful control

The anchor $q = (3 + 4i)/5$ lies on $U(1)$. Its quadratic trace is $6/5$, so it is not an algebraic integer and hence not a root of unity; $m \mapsto q^m$ is faithful on $\mathbb{Z}$. Compare the labelled triples $(-2, 0, 2)$ and $(-12, 0, 12)$. They are componentwise congruent modulo five, so every $\mathbb{Z}/5$-twisted trace and determinant agrees. At q, however, the linear coefficients of (3) are $-q^{-2}/2$ and $-q^{-12}/2$, which differ. The Laurent exponents recover both triples, and the same injectivity holds under exact evaluation at any known faithful character.

Progress over C129. C129 detected only translation residues modulo five. C134 removes that finite kernel while preserving its all-period cycles, global function space, and Fredholm owner. The gain is exact branch-labelled lattice recovery in the frozen scaled family, not complete geometric recovery.

5 Boundary and validation

The character coordinate must remain labelled. The precise sign relation is

$$D_{-t,u}(z) = D_{t,u^{-1}}(z),$$

which is parameter inversion, never a reciprocal determinant. Torsion-only samples retain finite kernels; exact injectivity supplies no finite-precision stability; and varying the graph, weights, linear part, branch labels, or noninteger geometry lies outside the theorem.

The exact receipt contains 284 rooted words, 40 primitive cycles, and all 12 permutation recoveries for $k = 1, 6$. A standard-library checker independent of the producer passes 64 assertions; an independent SymPy reconstruction passes 64 checks; byte replay and all 41 hostile cases (40 repaired-hash plus one stale-hash case) pass.

The strict tuple is (`A1_WEAK`, `A2_FAIL`, `A3_FAIL`, `A4_FORMAL_HINT`), overall `ROUTE_A_EXPLORATORY`. The character family is a formal phase lift, not a natural unitary, Hamiltonian, or metaplectic quantization. We claim no target divisor, prime-like correspondence, arithmetic/local data, Euler factors, root numbers, automorphy, Hilbert–Pólya operator, or Riemann-zero relation. Route B is unauthorized; the literal firewall is `NO_BAD_EULER_OR_ROOT_NUMBER`.